

# EAB 770 - Chapter 3

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## 1 Chapter 3: Sets of $2 \times 2$ tables

### 1.1 Outline

- Mantel-Haenszel test
- Measures of Association
  - Common Odds ratio
  - Breslow-Day test
  - Zelen's Test

## 2 Mantel-Haenszel Statistic

### 2.1 Sets of $2 \times 2$ tables

- Consider a multi-location RCT
  - Different locations will have different contingency tables
  - We want to consolidate **all** data and account for the location effect in testing for the association between the treatment and outcome.

**Table 3.1** Respiratory Improvement

| Center | Treatment | Yes | No | Total |
|--------|-----------|-----|----|-------|
| 1      | Test      | 29  | 16 | 45    |
| 1      | Placebo   | 14  | 31 | 45    |
| Total  |           | 43  | 47 | 90    |
| 2      | Test      | 37  | 8  | 45    |
| 2      | Placebo   | 24  | 21 | 45    |
| Total  |           | 61  | 29 | 90    |

## 2.2 Stratified Analysis

- Stratified analysis examines the association between two variables while adjusting for the effect of other covariates.
  - The variable that define the strata may be an explanatory variable, location, or classification variable.
  - Each table corresponds to one level of the explanatory variables (stratum)

## 2.3 Mantel-Haenszel Test

Recall the Mantel-Haenszel Chi-Square statistic

$$Q = (n_{11} - m_{11})^2 / v_{11}$$

where  $m_{11}$  is the expected value of the  $n_{11}$  and  $v_{11}$  is the variance of  $n_{11}$  based on the hypergeometric distribution assumption.

- If there are  $q$   $2 \times 2$  tables, each table will have a value for  $Q$ .

## 2.4 Mantel-Haenszel Test

The Mantel-Haenszel test assess the overall association of group and response after adjusting for the strata. If  $n_{j11}$  is the  $n_{11}$  value of the  $j^{th}$  contingency table, the Mantel-Haenszel statistic,  $Q_{MH}$  can be written as:

$$Q_{MH} = \frac{\left[ \sum_{j=1}^q n_{j11} - \sum_{j=1}^q m_{j11} \right]^2}{\sum_{j=1}^q v_{j11}}$$

**i** Note

$Q_{MH}$  is often called an **average partial association statistic**.

## 2.5 Mantel-Haenszel Statistic

- The statistic  $Q_{MH}$  relies on detecting patterns of the differences of proportions in each table
- The statistic  $Q_{MH}$  approximately follows a chi-square distribution with one degree of freedom if the combined random sample sizes are large.
  - Individual cell counts and isolated table sample sizes may be small.
  - The *Mantel-Fleiss criterion* can be used to check if the chi-square assumption holds. The MF score should be greater than 5.

## 2.6 Mantel-Haenszel Test: SAS

- The `cmh` option in the `tables` statement produces the Mantel-Haenszel statistic values. There will be three Mantel-Haenszel values included in the output.
- Nonzero Correlation
- Row Mean Scores Differ
- General Association

**i** Note

For  $2 \times 2$  tables, the scores and degrees of freedom do not differ for these three MH-statistics. These measures will be distinguished later in the course.

## 2.7 Mantel-Haenszel Test: SAS

- The `mf` option in the `tables` statement produces the Mantel-Fleiss criterion. Recall that the Mantel-Fleiss criterion (MF) should be greater than 5 for the chi-square distributional assumption to be valid.

## 2.8 Example

The table shows the results of a study about how general daily stress affects one's opinion on a proposed new health policy. Participants were sampled from urban and rural areas.

- Is there enough evidence that there is an association between the opinion and stress levels of residents after controlling for residential area? Use  $\alpha=0.05$  as the significance level.

**Table 3.3** Health Policy Opinion Data

| Residence | Stress | Favorable | Unfavorable | Total |
|-----------|--------|-----------|-------------|-------|
| Urban     | Low    | 48        | 12          | 60    |
| Urban     | High   | 96        | 94          | 190   |
|           | Total  | 144       | 106         | 250   |
| Rural     | Low    | 55        | 135         | 190   |
| Rural     | High   | 7         | 53          | 60    |
|           | Total  | 62        | 188         | 250   |

## 2.9 Example

The table shows the results of a study about how general daily stress affects one's opinion on a proposed new health policy. Participants were sampled from urban and rural areas.

- Is there enough evidence that there is an association between the opinion and stress levels of residents after controlling for residential area? Use  $\alpha=0.05$  as the significance level.

### 2.9.1 SAS Code

```
1 data stress;
2 input region $ stress $ outcome $ count @@;
3 datalines;
4 urban low f 48 urban low u 12
5 urban high f 96 urban high u 94
6 rural low f 55 rural low u 135
7 rural high f 7 rural high u 53
8 ;
9 proc freq order=data data=stress;
10 weight count;
11 tables region*stress*outcome / all chisq cmh (mf) nocol nopct;
12 run;
```

①

- ① CMH includes the Mantel-Haenszel statistics. MF gives the Mantel-Fleiss condition;

## 2.9.2 R Code

```
library(vcdExtra)
```

```
Loading required package: vcd
```

```
Loading required package: grid
```

```
Loading required package: gnm
```

```
Attaching package: 'vcdExtra'
```

```
The following object is masked from 'package:dplyr':
```

```
summarise
```

```
dat <- expand.grid(Outcome = c("Favorable","Unfavorable"),
                  Residence=c("Urban","Rural"),
                  Stress = c("Low","High")
                ) |>
mutate(count = c(48,12,55,135,96,94,7,53))

xtabs(count ~ Stress + Outcome + Residence ,data=dat) -> dat_cc

CMHtest(dat_cc,
        strata="Residence",
        types="ALL",
        overall=T)
```

```
$`Residence:Urban`
```

```
Cochran-Mantel-Haenszel Statistics for Stress by Outcome
in stratum Residence:Urban
```

|         | AltHypothesis          | Chisq  | Df | Prob       |
|---------|------------------------|--------|----|------------|
| general | General association    | 16.155 | 1  | 5.8367e-05 |
| rmeans  | Row mean scores differ | 16.155 | 1  | 5.8367e-05 |

```
cmeans Col mean scores differ 16.155 1 5.8367e-05
cor      Nonzero correlation 16.155 1 5.8367e-05
```

```
$`Residence:Rural`
Cochran-Mantel-Haenszel Statistics for Stress by Outcome
  in stratum Residence:Rural
```

|         | AltHypothesis          | Chisq  | Df | Prob      |
|---------|------------------------|--------|----|-----------|
| general | General association    | 7.2724 | 1  | 0.0070022 |
| rmeans  | Row mean scores differ | 7.2724 | 1  | 0.0070022 |
| cmeans  | Col mean scores differ | 7.2724 | 1  | 0.0070022 |
| cor     | Nonzero correlation    | 7.2724 | 1  | 0.0070022 |

```
$ALL
Cochran-Mantel-Haenszel Statistics for Stress by Outcome
  Overall tests, controlling for all strata
```

|         | AltHypothesis          | Chisq | Df | Prob       |
|---------|------------------------|-------|----|------------|
| general | General association    | 23.05 | 1  | 1.5783e-06 |
| rmeans  | Row mean scores differ | 23.05 | 1  | 1.5783e-06 |
| cmeans  | Col mean scores differ | 23.05 | 1  | 1.5783e-06 |
| cor     | Nonzero correlation    | 23.05 | 1  | 1.5783e-06 |

### 2.9.3 Answer

$\chi^2 = 23.05$ ,  $df = 1$ ,  $p < 0.0001$ . There is strong evidence that the stress level is associated with the opinion outcome.

## 2.10 Example

The table shows the results of an RCT for a test treatment designed for respiratory infections. Is there a significant association between the treatment and the outcome after controlling for the center?

- Is the Mantel-Fleiss criterion satisfied?

```
, , Center = 1
```

```
Improvement
```

| Treatment | Yes | No |
|-----------|-----|----|
| Test      | 29  | 16 |
| Placebo   | 14  | 31 |

, , Center = 2

|           | Improvement |    |
|-----------|-------------|----|
| Treatment | Yes         | No |
| Test      | 37          | 8  |
| Placebo   | 24          | 21 |

## 2.11 Example

The table shows the results of an RCT for a test treatment designed for respiratory infections. Is there a significant association between the treatment and the outcome after controlling for the center?

- Is the Mantel-Fleiss criterion satisfied?

### 2.11.1 SAS Code

```
data resp;
length treatment $7.;
input treatment $ improvement $ center $ count;
datalines;
Test Yes 1 29
Placebo Yes 1 14
Test No 1 16
Placebo No 1 31
Test Yes 2 37
Placebo Yes 2 24
Test No 2 8
Placebo No 2 21
;

proc freq data=resp order=data;
weight count;
tables center*treatment*improvement / cmh (MF) nocol norow nopct;
run;
```

### 2.11.2 R Code

```
dat <- expand.grid(Improvement = c("Yes","No"),
                  Center=c(1,2),
                  Treatment = c("Test","Placebo")
                  ) |>
mutate(count = c(29,16,37,8,14,31,24,21))

xtabs(count ~ Treatment+Improvement + Center, data=dat) -> dat_cc
dat_cc
```

```
, , Center = 1
```

|           | Improvement |    |
|-----------|-------------|----|
| Treatment | Yes         | No |
| Test      | 29          | 16 |
| Placebo   | 14          | 31 |

```
, , Center = 2
```

|           | Improvement |    |
|-----------|-------------|----|
| Treatment | Yes         | No |
| Test      | 37          | 8  |
| Placebo   | 24          | 21 |

```
CMHtest(dat_cc,
        strata="Center",
        types="ALL",
        overall=T)
```

```
$`Center:1`
```

```
Cochran-Mantel-Haenszel Statistics for Treatment by Improvement
in stratum Center:1
```

|         | AltHypothesis          | Chisq  | Df | Prob      |
|---------|------------------------|--------|----|-----------|
| general | General association    | 9.9085 | 1  | 0.0016452 |
| rmeans  | Row mean scores differ | 9.9085 | 1  | 0.0016452 |
| cmeans  | Col mean scores differ | 9.9085 | 1  | 0.0016452 |
| cor     | Nonzero correlation    | 9.9085 | 1  | 0.0016452 |



```
$`Center:2`
```

```
Cochran-Mantel-Haenszel Statistics for Treatment by Improvement  
in stratum Center:2
```

|         | AltHypothesis          | Chisq  | Df | Prob      |
|---------|------------------------|--------|----|-----------|
| general | General association    | 8.5025 | 1  | 0.0035465 |
| rmeans  | Row mean scores differ | 8.5025 | 1  | 0.0035465 |
| cmeans  | Col mean scores differ | 8.5025 | 1  | 0.0035465 |
| cor     | Nonzero correlation    | 8.5025 | 1  | 0.0035465 |

```
$ALL
```

```
Cochran-Mantel-Haenszel Statistics for Treatment by Improvement  
Overall tests, controlling for all strata
```

|         | AltHypothesis          | Chisq  | Df | Prob       |
|---------|------------------------|--------|----|------------|
| general | General association    | 18.411 | 1  | 1.7807e-05 |
| rmeans  | Row mean scores differ | 18.411 | 1  | 1.7807e-05 |
| cmeans  | Col mean scores differ | 18.411 | 1  | 1.7807e-05 |
| cor     | Nonzero correlation    | 18.411 | 1  | 1.7807e-05 |

### 2.11.3 Answer

- The Mantel-Fleiss criterion was determined to be 36, which means the chi-square assumption holds.
- $Q_{MH} = 18.4$ ,  $p < 0.0001$ , there is strong evidence of an association between the treatment and improvement.

## 3 Measures of Association

### 3.1 Common Odds Ratio

- An odds ratio can be calculated for each stratum based on the 2x2 table values.
  - What if there are multiple strata?
- If the odds ratios are expected to be the same over all strata (homogenous), then we can calculate a common odds ratio  $\psi$ .

## 3.2 Common Odds Ratio

- The Mantel-Haenszel estimator  $\hat{\psi}_{MH}$  can approximate the common odds ratio such that

$$\hat{\psi}_{MH} = \frac{\sum_{j=1}^q n_{j11}n_{j22}/n_j}{\sum_{j=1}^q n_{j12}n_{j21}/n_j}$$

### Note

We can calculate confidence intervals (asymptotic and exact) for common odds ratios!

## 3.3 Common Odds Ratio: Confidence Intervals

There are two confidence intervals for the common odds ratios provided by SAS: The Mantel-Haenszel and Logit confidence limits.

### 3.3.1 Mantel-Haenszel

- based on an estimate of the variance of  $\log \psi_{MH}$
- Not sensitive to small sample sizes

### 3.3.2 Logit

- based on weighted regression estimates
- reasonable but requires adequate sample sizes, i.e.  $n_{hij} \geq 5$

## 3.4 Homogeneity of Odds Ratios

### Warning

It is important to remember that the common odds ratio assumes the odds ratio is uniform across all strata!

The Breslow-Day test or the Zelen's test can be used to test if the homogeneity assumption is satisfied.

### 3.4.1 Breslow-Day test

- Null hypothesis: All strata are homogeneous.
- Assume the null value of the odds ratio is  $\psi_{MH}$  for each stratum
- Calculate the expected values of the cells based on  $\psi_{MH}$

$$Q_{BD} = \sum_{h=1}^q \sum_{i=1}^q \sum_{j=1}^q \frac{(n_{hij} - m_{hij})^2}{m_{hij}}$$

which approximately follows a chi-square distribution with  $(n - 1)$  degrees of freedom.

### 3.4.2 Zelen's Test

Zelen's test is similar to the Breslow-Day test, but uses exact methods.

- Recall: Fisher's Exact Test
- calculates the probability of all possible sets of tables.
- p-value is the sum of all table probabilities that are lower than the probability of the observed table.

## 3.5 Homogeneity of Odds ratios

If the Breslow-Day test or Zelen's test is significant, then the common odds ratio might not be an appropriate measure of association.

- The strata must be analyzed separately (similar to interactions in linear models.)

## 3.6 Common Odds Ratios: SAS

- The common odds ratios are calculated with the `cmh` option
  - Only includes asymptotic (MH and logit) CI limits.
  - Including the `measures` options yields plots of common odds ratios and stratum-level odds ratios.
- The Breslow-Day test is also included in the `cmh` option
- The `comor` and `eqor` options in the `exact` statement yield the exact common odds ratio and the Zelen's test results, respectively.

Sample SAS code:

```

1 proc freq;
2 tables a*b*c/cmh measures;
3 exact comor eqor;
4 run;

```

### 3.7 Example

The table shows the data from a study on coronary artery disease. Investigators were interested in whether electrocardiogram (ECG) measurement was associated with disease status. Gender was thought to be associated with disease status, so investigators post-stratified the data into male and female groups. In addition, there was interest in examining the odds ratios.

**Table 3.7** Coronary Artery Disease Data

| Sex    | ECG                         | No Disease | Disease | Total |
|--------|-----------------------------|------------|---------|-------|
| Female | < 0.1 ST segment depression | 11         | 4       | 15    |
| Female | ≥ 0.1 ST segment depression | 10         | 8       | 18    |
| Male   | < 0.1 ST segment depression | 9          | 9       | 18    |
| Male   | ≥ 0.1 ST segment depression | 6          | 21      | 27    |

### 3.8 Example

The table shows the data from a study on coronary artery disease. Investigators were interested in whether electrocardiogram (ECG) measurement was associated with disease status. Gender was thought to be associated with disease status, so investigators post-stratified the data into male and female groups. In addition, there was interest in examining the odds ratios.

#### 3.8.1 SAS Code

```

data ca;
input sex $ ECG $ disease $ count;
datalines;
female <0.1 yes 4
female <0.1 no 11
female >=0.1 yes 8
female >=0.1 no 10
male <0.1 yes 9
male <0.1 no 9
male >=0.1 yes 21
male >=0.1 no 6

```

```

;
proc freq data=ca ;
weight count;
tables sex*disease / nocol nopct chisq measures;
tables sex*ECG*disease / nocol nopct cmh measures;
run;

```

- ① Testing for an association between sex and disease status
- ② Using sex as a stratification variable/strata. **measures** includes common odds ratio plots.

### 3.8.2 Answer

- Sex was determined to be associated with disease status ( $\chi^2 = 6.9, p = 0.0084$ )
- Controlling for sex, there is strong evidence that the ECG measurements are associated with disease status ( $\chi^2 = 4.5, p = 0.03$ ).
- The Breslow-Day test yielded a non-significant p-value ( $p=0.64$ ), hence we can use the common odds ratios
- The MH odds ratio was estimated to be 2.86 (95% CI: 1.1,7.5). We can interpret this value as: The odds of having the disease is 2.86 times higher for patients with a high ECG segment depression compared to those with a low ECG segment depression.

### 3.9 Exercise

Table 3.8 shows the data from research done by a small company in initiating exercise programs (office/home) in their two locations in a nearby suburbs. After a year, participants and non-participants underwent a cardiovascular stress test to assess their fitness, and their result was recorded as good or not good depending on age-adjusted criteria. The exercise counselor was interested in whether type of program was associated with good test results.

- Calculate the common odds ratio  $\psi_{MH}$  and the **exact** 95% confidence intervals.
- Use Zelen's test to test if the odds ratio is uniform in both locations.

**Table 3.8** Cardiovascular Test Outcomes

| Location  | Program | Good | Not Good | Total |
|-----------|---------|------|----------|-------|
| Downtown  | Office  | 12   | 5        | 17    |
| Downtown  | Home    | 3    | 5        | 8     |
|           | Total   | 15   | 10       | 25    |
| Satellite | Office  | 6    | 1        | 7     |
| Satellite | Home    | 1    | 3        | 4     |
|           | Total   | 7    | 4        | 11    |

### 3.10 Exercise

Table 3.8 shows the data from research done by a small company in initiating exercise programs (office/home) in their two locations in a nearby suburbs. After a year, participants and non-participants underwent a cardiovascular stress test to assess their fitness, and their result was recorded as good or not good depending on age-adjusted criteria. The exercise counselor was interested in whether type of program was associated with good test results.

- Calculate the common odds ratio  $\psi_{MH}$  and the **exact** 95% confidence intervals.
- Use Zelen's test to test if the odds ratio is uniform in both locations

#### 3.10.1 SAS Code

```
data exercise;
input location $ program $ outcome $ count @@;
datalines;
downtown office good 12 downtown office not 5
downtown home good 3 downtown home not 5
satellite office good 6 satellite office not 1
satellite home good 1 satellite home not 3
;

proc freq order=data;
weight count;
tables location*program*outcome /cmh(mf);
exact comor eqor;
run;
```

①

②

- ① MF should be enclosed in parentheses
- ② `eqor`: Zelen's test, `comor`: exact confidence interval for the common odds ratio.

### 3.10.2 Answer

- Mantel-Fleiss criterion is not satisfied (MF=4.65), so be careful with using the asymptotic chi-square test for  $Q_{MH}$ .
  - This motivates using exact methods in calculating effect sizes.
- Common odds ratio was estimated to be 5.84 (**EXACT** 95% CI: 1.05, 33.3). There is sufficient evidence at the 0.05 significance level that the program type is associated with the quality of cardiovascular test results.
  - The Zelen's test yielded  $p = 1$ , which means there is no evidence against homogeneity.